Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: LTZGLUE | Context: Luxembourgish Professional Newsroom

Definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Justification

Modality alignment is good: both benchmark and deployment are text-only [Q17], RTL articles, comments, parliamentary transcripts, and chat rooms provide register diversity in pre-training [Q74], and OpenLID filtering removes non-Luxembourgish content [Q47]. The 40–400 word filter [Q47] is plausibly compatible with newsroom article lengths. However, Luxembourgish has 'ongoing standardisation' and 'vast amounts of variation' [Q7], and the documentation does not describe explicit orthographic normalisation across datasets. The BPE tokenizer (50,368 vocab on 233M tokens) [Q128, Q75] may not adequately represent journalistic orthographic variants or code-switched French/German tokens, particularly given the formal-register skew of the data [Q112]. The deployment explicitly applies the flexible journalistic register [A3], but LA Ortho ground truth is built from Spellchecker.lu data [Q31] encoding the formal ZLS standard [WEB-11, WEB-12], creating a known register mismatch.

ID	Evidence
WEB-11	LA Ortho is anchored in Spellchecker.lu which encodes the formal ZLS standard, structurally divorced from the journalistic register the deployment applies

Strengths

- Text-only modality matches the deployment's text-only pipeline [Q17]
- Pre-training corpus draws from a broad set of Luxembourgish written sources including RTL news, comments, parliamentary speech, podcasts, chat rooms, and Wikipedia [Q74], providing some register breadth
- Article-length filtering (40–400 words) [Q47] is operationally compatible with newsroom article lengths

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Both benchmark and deployment use plain text [Q17]; signal-distribution mismatch is low at the modality level. Tokenisation (BPE, 50,368 vocab on 233M tokens) [Q128, Q75] may under-represent journalistic orthographic variants and code-switched tokens.

IF-2: Luxembourg has near-universal digital infrastructure (98.8% internet penetration [WEB-17]), so capture infrastructure is not a constraint.

ID	Evidence
WEB-17	98.8% internet penetration in Luxembourg confirms enterprise-grade digital infrastructure for newsroom deployment

IF-3: The benchmark filters articles to 40–400 words [Q47] and removes non-Luxembourgish content via OpenLID [Q47] — both operationally plausible for newsroom text. However, no explicit orthographic normalisation across datasets is documented despite acknowledged 'vast amounts of variation' [Q7], and the LA Ortho category encodes the ZLS formal standard via Spellchecker.lu [Q31, WEB-11] which diverges from the deployment's flexible journalistic register [A3].

ID	Evidence
WEB-11	LA Ortho is anchored in Spellchecker.lu which encodes the formal ZLS standard, structurally divorced from the journalistic register the deployment applies

IF-4: Form-level mismatches that may harm external validity: (a) tokenisation potentially mis-segmenting code-switched and orthographically variant tokens; (b) implicit assumption of formal orthography in LA Ortho ground truth diverging from journalistic practice [A3, Q31].

Information Gaps

- Whether explicit orthographic normalisation was applied across datasets (gap_id 3, partially resolved — formal-source inputs confirmed but normalisation across datasets undocumented)
- Tokenizer behaviour on code-switched and orthographically variant tokens

Requires Expert Verification

- Whether the BPE tokenizer adequately segments code-switched newsroom copy

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Remediation

Gap 1: No explicit orthographic normalisation across datasets despite acknowledged 'vast amounts of variation' [Q7]; tokenizer behaviour on code-switched and orthographically variant tokens is undocumented [Q128]

Recommendation: Profile the BPE tokenizer's handling of code-switched French/German tokens and journalistic orthographic variants on a sample of newsroom copy; consider tokenizer extension or fine-tuning if fragmentation rates are high